



Product Training

# Agisoft Photogrammetry

Version: B

Proprietary & Confidential



WELCOME

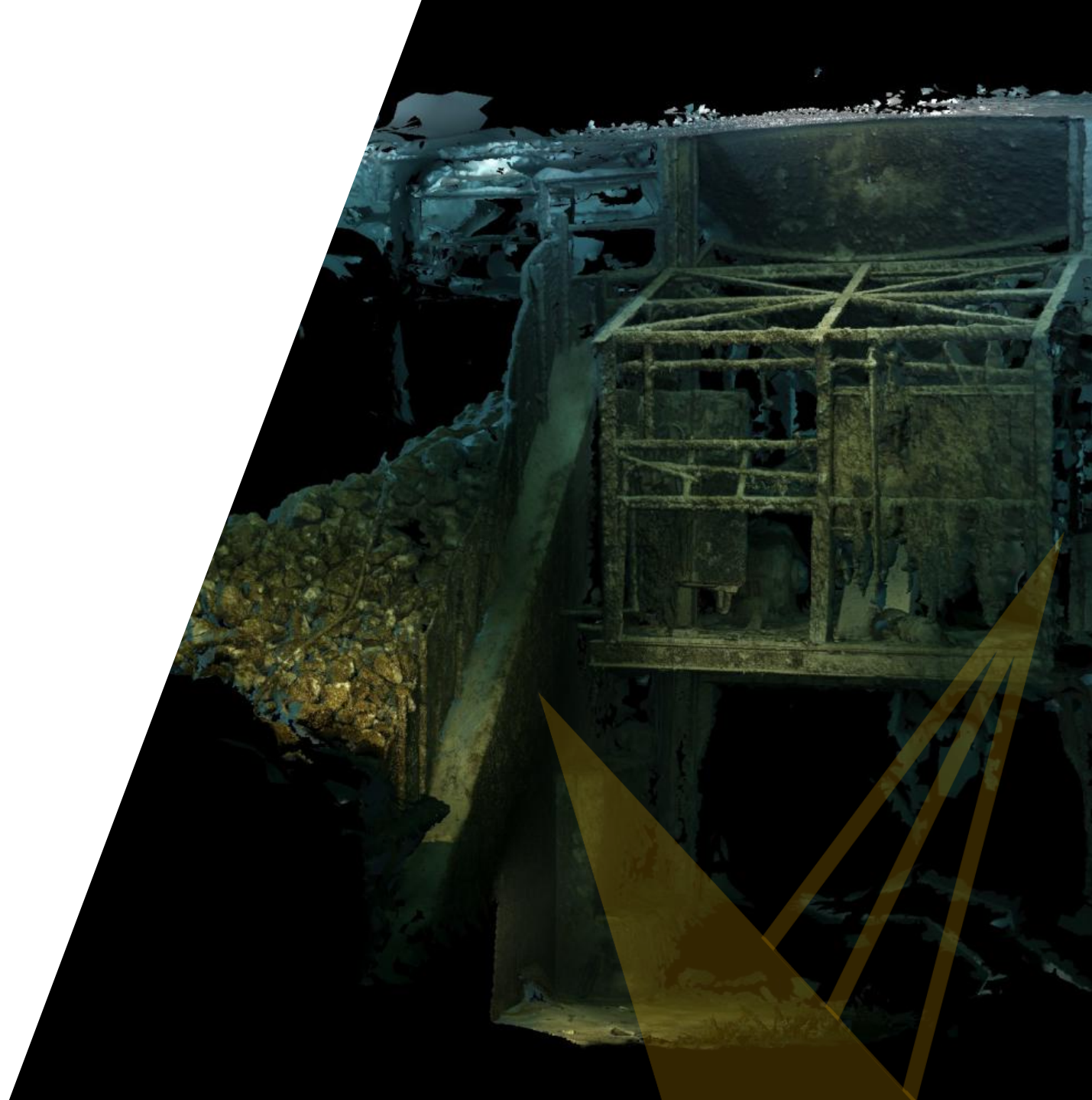
# Agenda

1. Overview
2. Prerequisites
3. Agisoft Setup
4. Align & Reduce
5. Build Mesh
6. Texture
7. Animate
8. Exports



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Voyis Technical Support



# Overview





OVERVIEW

# What is Agisoft?



Agisoft is a stand-alone software product that performs photogrammetric processing of input digital stills images and generates 3D spatial data to be used in survey applications.

## Why use Agisoft with Voyis Products?

Voyis camera products are capable of capturing still images underwater with unparalleled quality compared to other products in its class. With Agisoft's advanced technology, you can transform these **monocular or stereo still images** into stunning 3D models.

In this presentation, you will be guided on how to create your first model in Agisoft. If you need further training in Agisoft, we suggest buying a training course from a reliable source.

**This guide explains the exact workflow for using Voyis images in Agisoft software to generate accurate 3D models.**

**It focuses on a workflow for stereo-images from the [Discovery Stereo](#) product, but process is mostly the same for all of Voyis' monocular stills camera products.**



## OVERVIEW

# Mono vs Stereo Photogrammetry

## Monocular Cameras



Discovery



Observer



Insight  
Micro



Recon

- A sequence of Stills Images from a single camera triggered at a constant rate (1 to 5 fps)
- Input: Voyis calibrated stills images (processed)
- Calibrated images have a camera calibration to undistort images for accurate reconstruction
- **3D Modelling:** The resulting 3D model has no scale (dimensionless) unless scale-bars or navigation data is input as well

## Stereo Camera



Discovery  
Stereo

- A sequence of Stereo-Pair images from two cameras triggered at the same time
- Input: Voyis calibrated stills image pairs (processed)
- Calibrated images have a camera calibration, and stereo calibration (Offset between the cameras)
- **3D Modelling:** The resulting 3D model is fully scaled without any additional input data

# Prerequisites







## PREREQUISITES

# Stills Image Format

This guide assumes that the input images are saved in the Voyis “Processed” format with “Undistortion” enabled. This ensures the input images are “Calibrated” and “Corrected”, ready for photogrammetry.

- **Calibrated** – Undistorted & Rectified (stereo)
  - Images will have “\_CAL” in file name for Discovery Products
- **Image Corrections** – Light levelling, colour correction
  - Images will have “\_processed” in file name
- Voyis has validated the system accuracy using its Calibrated Images with its measured calibration parameters.
- **Note:** Raw uncalibrated images can be used, but require extra camera model distortion parameters as inputs.



**Filename:** Image\_left\_processed\_SYSTEM\_2023-06-29T173731.825465\_CAL\_918.jpg

**Image Pair ID**

Stereo Only

**Corrected Image**

8-bit .JPG; RAW is alternative

**Time Source**

System or External

**“Date” T “Time”**

System OR PPS,PTP,NTP

**Calibrated Image**

Discovery Products only

**Image #**

Image in sequence

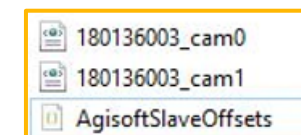


## PREREQUISITES

# Calibration Information

Voyis provides three “Calibration Files” with its imaging systems. These text files have the calibration parameters formatted for input to Agisoft. These are only valid for the associated product.

- **Camera Calibration files**, for one or both cameras (cam0, cam1) – **Can be directly loaded into Agisoft!**
- **Offset Calibration file**, defining the relative position of each camera – **Manually input into Agisoft!**
  - **Note:** Offset Calibration file is only applicable for the Discovery Stereo. Omit for monocular products.



**Camera calibration files** have **Rectified** parameters for **f**, **cx**, and **cy** valid for Calibrated Voyis Images:

| Camera Parameter | Description                                                                                                       | Typical Convention                                      | Agisoft Convention       | Files provided already have this conversion completed!                                   |
|------------------|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|--------------------------|------------------------------------------------------------------------------------------|
| f                | Focal length measured in pixels (derived from fx, fy of our pre-rectified model.) f is the same for both cameras. | Units of Pixels                                         | Units of Pixels          | 1:1                                                                                      |
| CX               | Principal point coordinates, i.e. coordinates of lens optical axis interception with sensor plane in pixels       | Zero is Top Left Corner of Top Left Pixel. +x is right. | Zero is center of sensor | $Cx\_Agisoft = Cx\_Voyis - 2816/2$<br>$Cx\_Voyis = 1438.926$<br>$Cx\_Agisoft = 30.926$   |
| cy               | Principal point coordinates, i.e. coordinates of lens optical axis interception with sensor plane in pixels       | Zero is Top Left Corner of Top Left Pixel. +y is down.  | Zero is center of sensor | $Cy\_Agisoft = Cy\_Voyis - 2816/2$<br>$Cy\_Voyis = 1402.8915$<br>$Cy\_Agisoft = -5.1085$ |

*Table for advanced users only. Information not required.*





## PREREQUISITES

# Calibration Information

**Offset Calibration file – Discovery Stereo** - Position & Orientation of the Right camera [**Right slave**] with respect to the left

- Parameters Translation [**x, y, z**], & Rotation [**Omega, Phi, Kappa**]
- These parameters must be copied from the file, and pasted into the Agisoft fields (manual input)
- **Note:** Voyis Calibrated Images have already been rectified, therefore “X” is the only non-zero value.

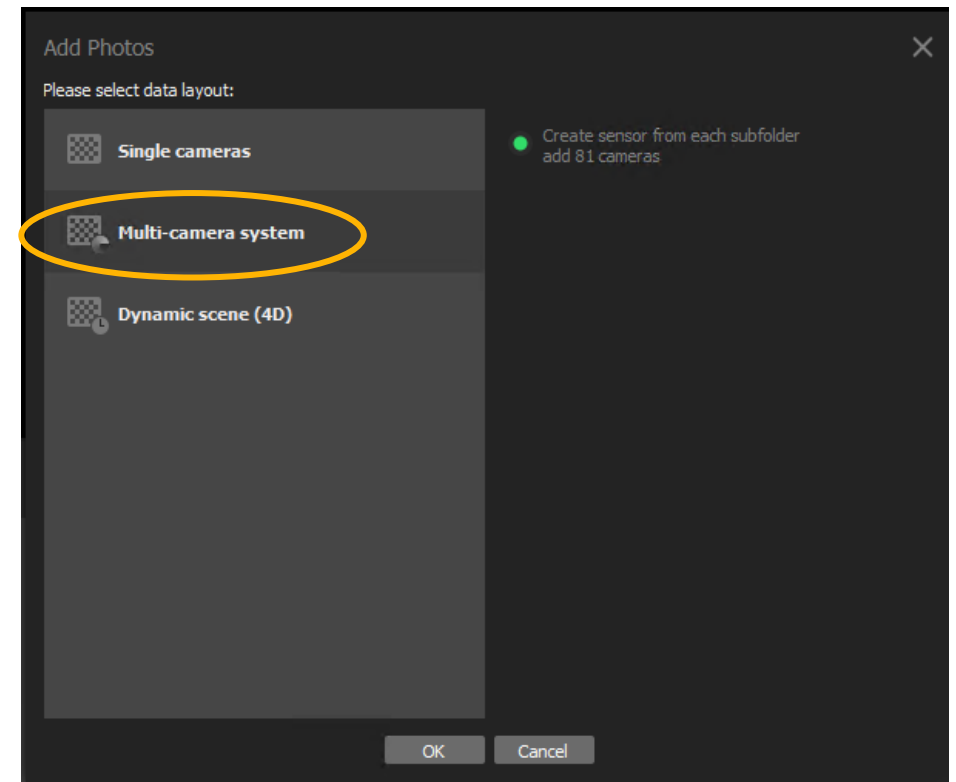
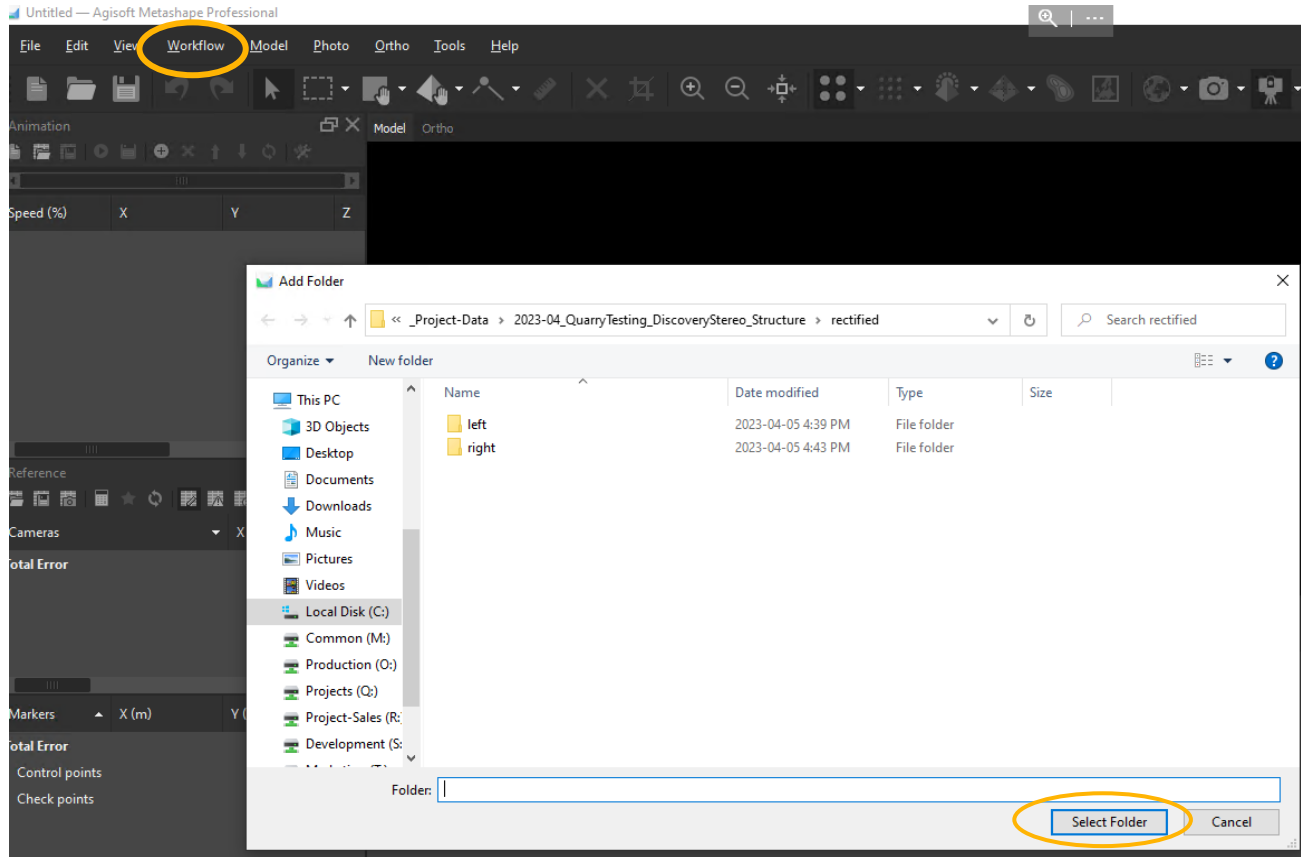
| Offset Parameter  | Description                                                   | Agisoft Convention                     |
|-------------------|---------------------------------------------------------------|----------------------------------------|
| Translation x,y,z | Position of the right camera, relative to the left camera.    | meters<br>[X, Y, Z]                    |
| Rotation          | Orientation of the right camera, relative to the left camera. | DEGREES (EULER)<br>[Omega, Phi, Kappa] |

# Agisoft Setup



# Input Stills Images

1. Open Root Camera Folder from the Workflow menu: Workflow → **Add Folder**
  - Select the Top-Level folder containing both Right and Left folders
2. Select “**Multi-camera system**” for the Discovery Stereo (image pairs) or “**single cameras**” for monocular products
3. Save a MetaShape project: File → Save





## Calibration Parameter Input

To simplify the Agisoft workflow, each imaging product will be accompanied by a USB stick containing the necessary calibration files.

- In case you need a backup, Voyis keeps these files saved.
- If needed, contact the Voyis Customer Support team with your product serial number

***This is an example of the files located on with Stereo Camera USB stick***

**Camera Calibration 1**  
(Left Camera for Discovery Stereo)

**Camera Calibration 2**  
(Right Camera for Discovery Stereo)  
(Second Calibration file only for Stereo)

**Agisoft Slave Offsets**  
(Discovery Stereo Only)

| PC > Local Disk (C:) > Datasets > 180136003 > AgisoftParams                                           |                     |                  |      |  |
|-------------------------------------------------------------------------------------------------------|---------------------|------------------|------|--|
| Name                                                                                                  | Date modified       | Type             | Size |  |
|  180136003_cam0      | 2023-07-20 12:20 PM | XML Document     | 1 KB |  |
|  180136003_cam1      | 2023-07-20 12:20 PM | XML Document     | 1 KB |  |
|  AgisoftSlaveOffsets | 2023-07-20 12:20 PM | JSON Source File | 1 KB |  |



# Add Calibration Files

## 1. Go to **Tools** → **Camera Calibration**

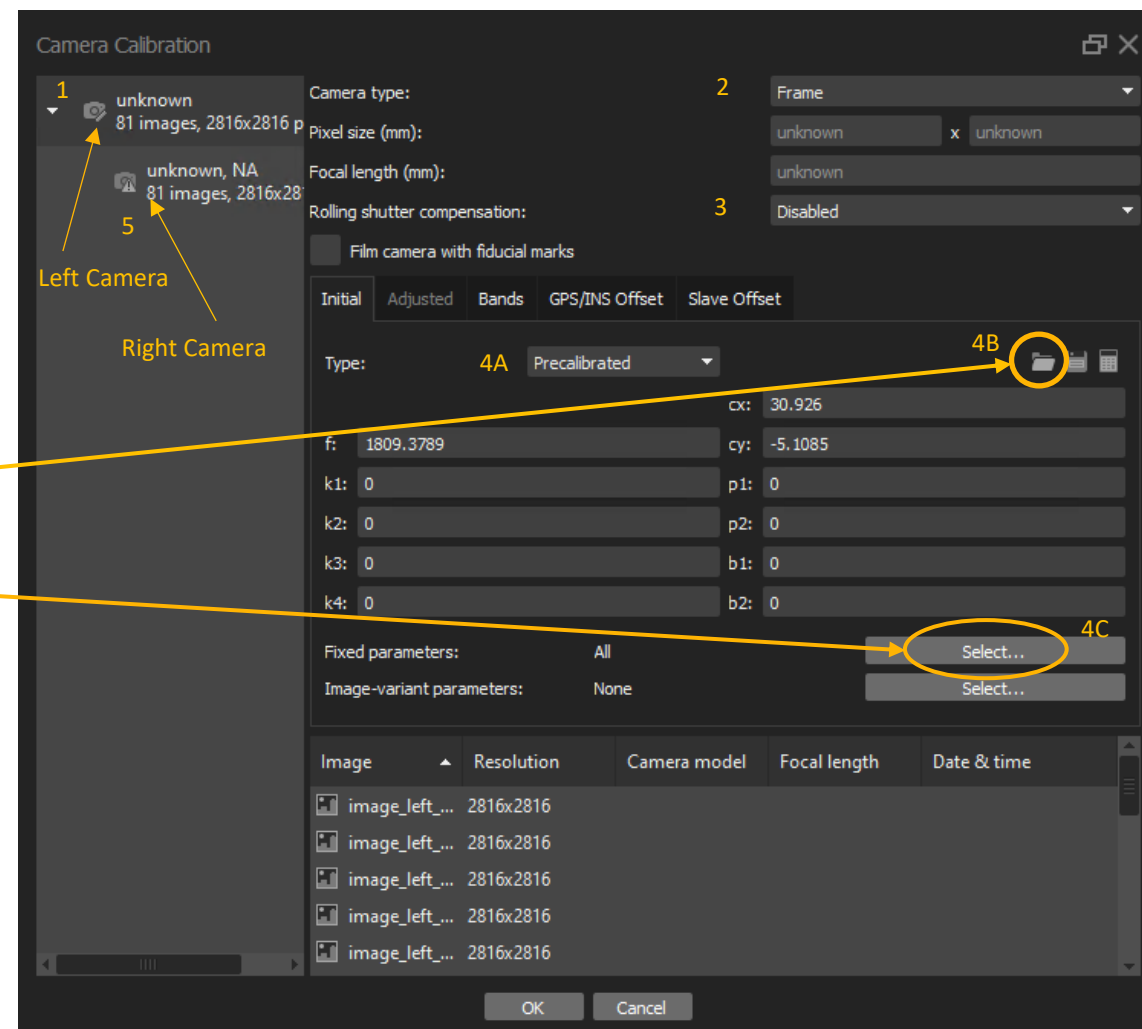
- A. In the tree you will see a root camera which is the Left Camera and a “sub-camera” which is the Right Camera. Start with the **LEFT** Camera.

## 2. Camera type = **Frame**

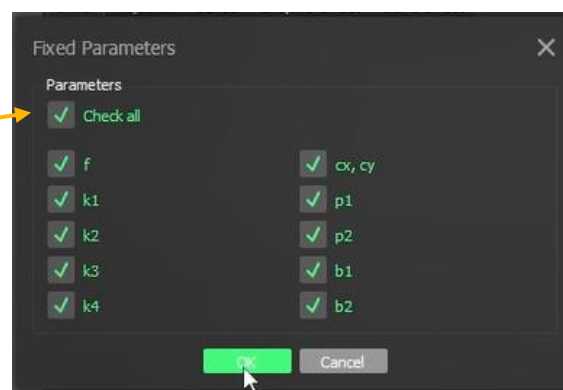
## 3. Rolling Shutter compensation = **Disabled**

## 4. Under the “**Initial**” Tab

- A. Type = **Precalibrated**
- B. Load camera calibration file to input **f**, **Cx**, and **Cy** parameters automatically. Click on the folder icon to load your calibration file for the Left camera (cam0)
- C. Fixed Parameters: Click on “**Select**”.



Select all “Check all”



5. REPEAT 1-4 for the **RIGHT** Camera (cam1 from your folder) if you are using the Discovery Stereo product.

**Note:** Only 1 camera will be shown for single camera systems





# Add Offset Calibration

## 1. In **Tools** → **Camera Calibration**

- Select the **Right** (sub) Camera.

## 2. Under “**Slave Offset**” Tab

- X,Y,Z Section:
  - ☒ Select - Enable Reference
  - ☒ Select - Adjust Location
- Manually input the Extrinsic Translation Value for “**X**”. The number will come from **Agisoft Slave Offsets**. The **Y** and **Z** values will always be **0**

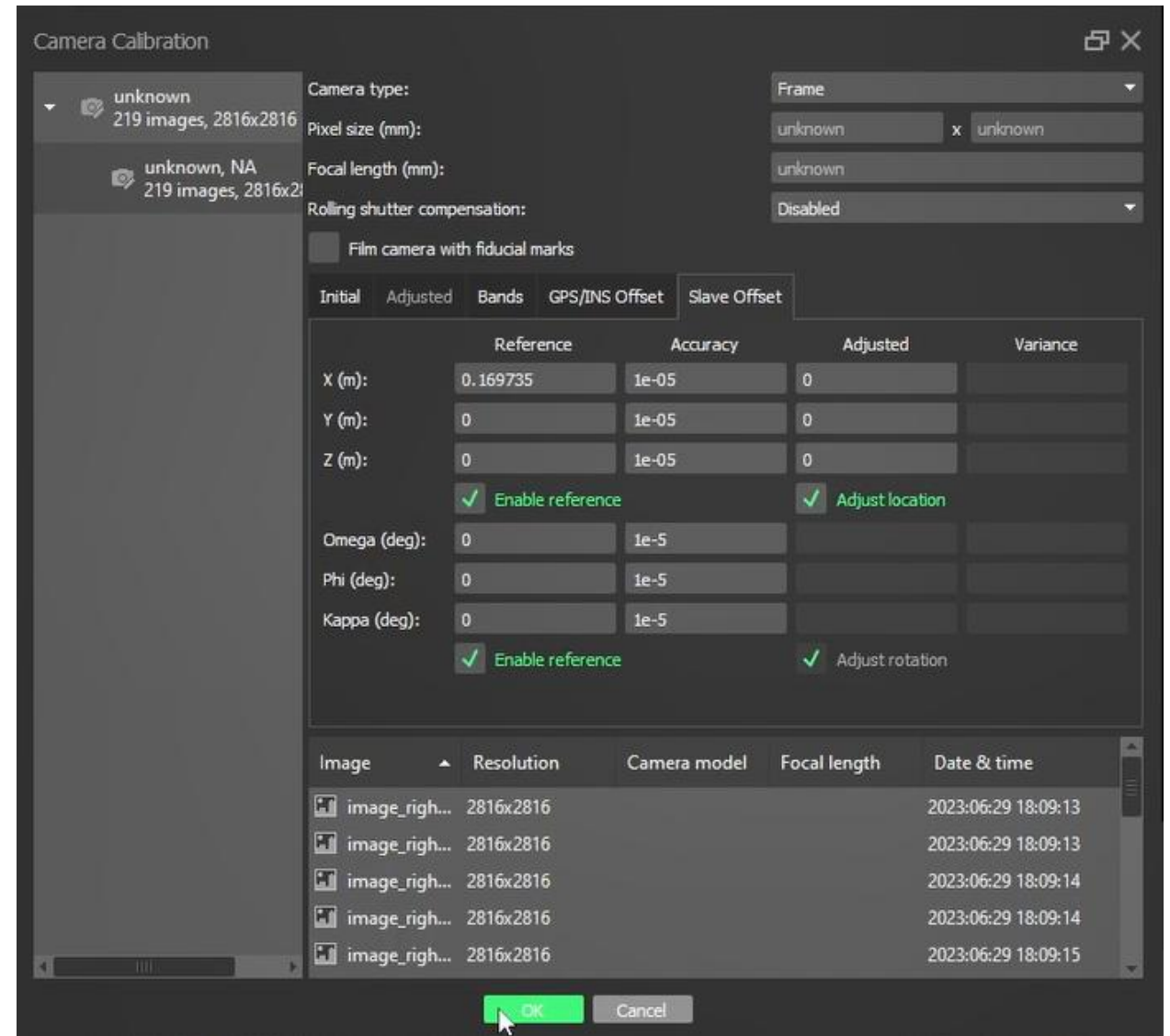
```
C: > Datasets > 180136003 > AgisoftParams > {} AgisoftSlaveOffsets.json > ...
1  {
2    "x": 0.16999040846861818,
3    "y": 0,
4    "z": 0,
5    "Omega": 0,
6    "Kappa": 0,
7    "Phi": 0
8  }
```

- Set Accuracy **1e-5** for all three fields

## 3. Omega, Phi, Kappa Section:

- ☒ Enable reference
- ☒ Adjust Location
- Manually input **0** for each field
- Set Accuracies **1e-5** (max) for all three fields

## 3. Click **OK**



**Note:** Only applicable to the Discovery Stereo product

# Align & Filter





# Align Settings

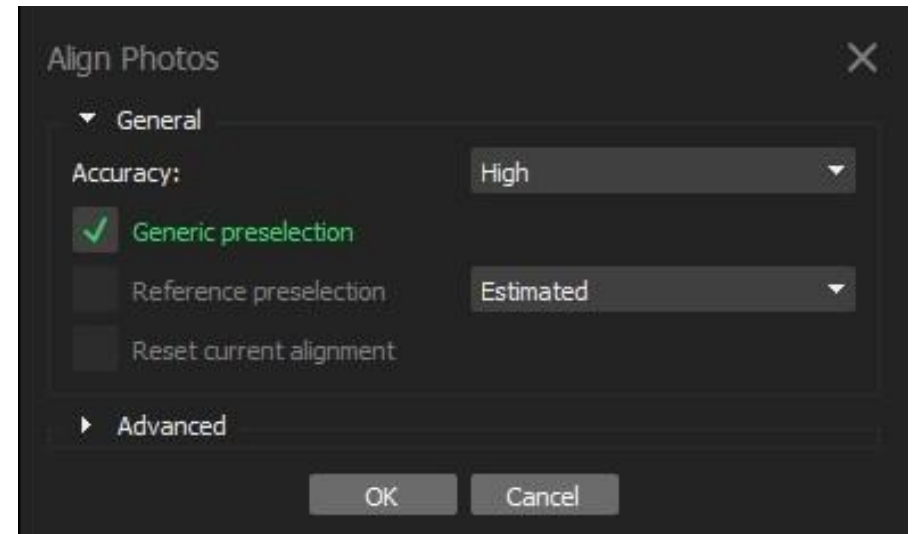
This portion of the workflow produces an initial sparse point cloud and determines the camera trajectory. It will also filter out low quality data and continuously optimize the solution.

## 1. Workflow → Align Photos

- Accuracy = **High**
  - High uses original images (no upscale/downscale)
  - Low can be used to downscale to speed up processing
- ONLY Click “Reference Preselection” → Select: **Estimated**.  
**If you know your data was collected in a single pass.**
  - Can also Click “Generic Preselection” but it takes longer to process and often doesn’t improve model

## 2. Click “OK” and wait for processing

## 3. Ensure the resulting sparse point cloud is representative of your object with no strange skews or bends





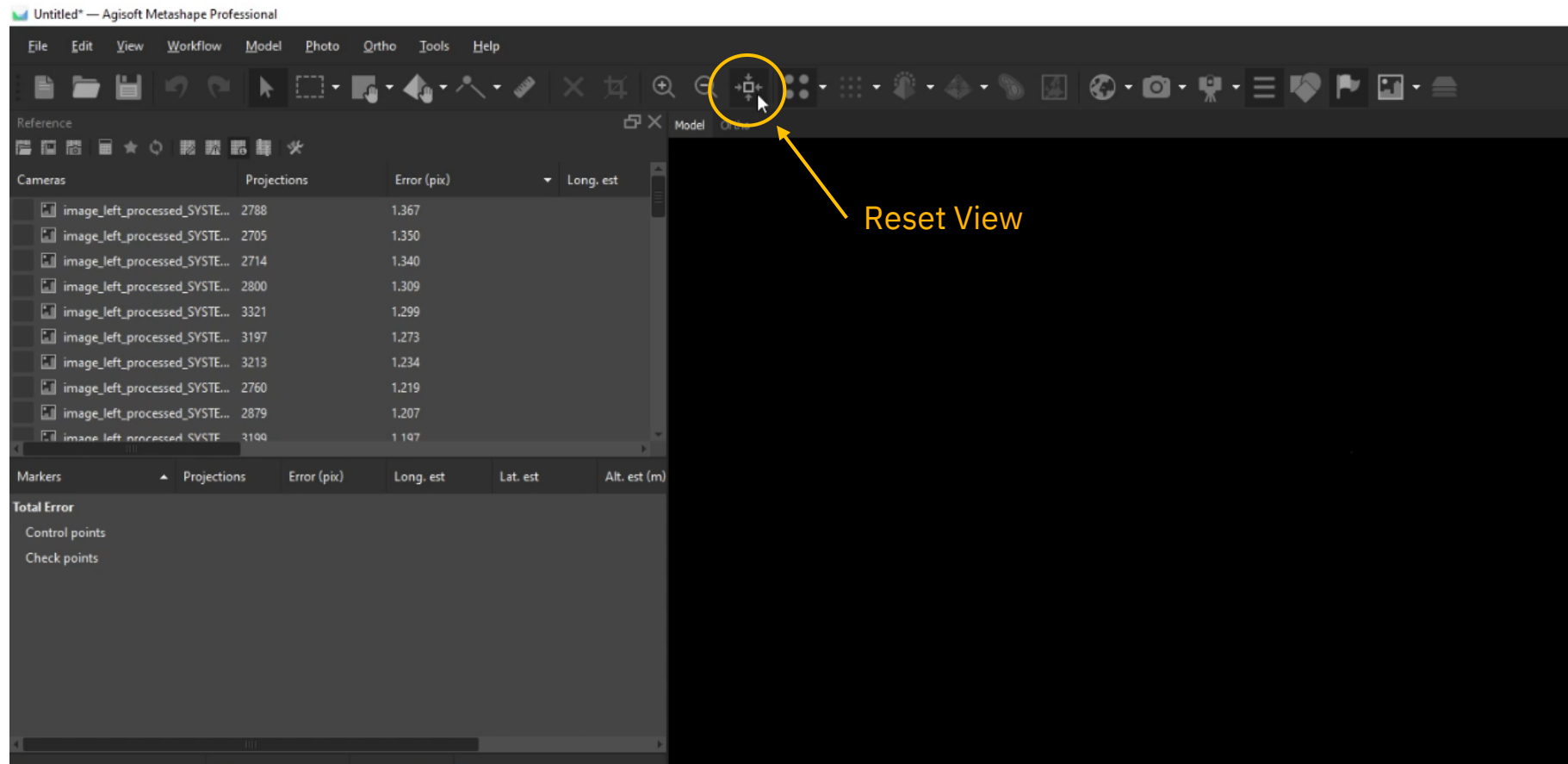
# Align Settings

**Remember, not all images will always align. This is often dictated by multiple factors in which the pilot of the ROV must be aware of. Below are just a few examples to keep in mind while capturing images for your 3D model:**

- Speed of the ROV
- Keep the object in view
- Water turbidity
- Capture 70% - 80% overlap

# Reset View

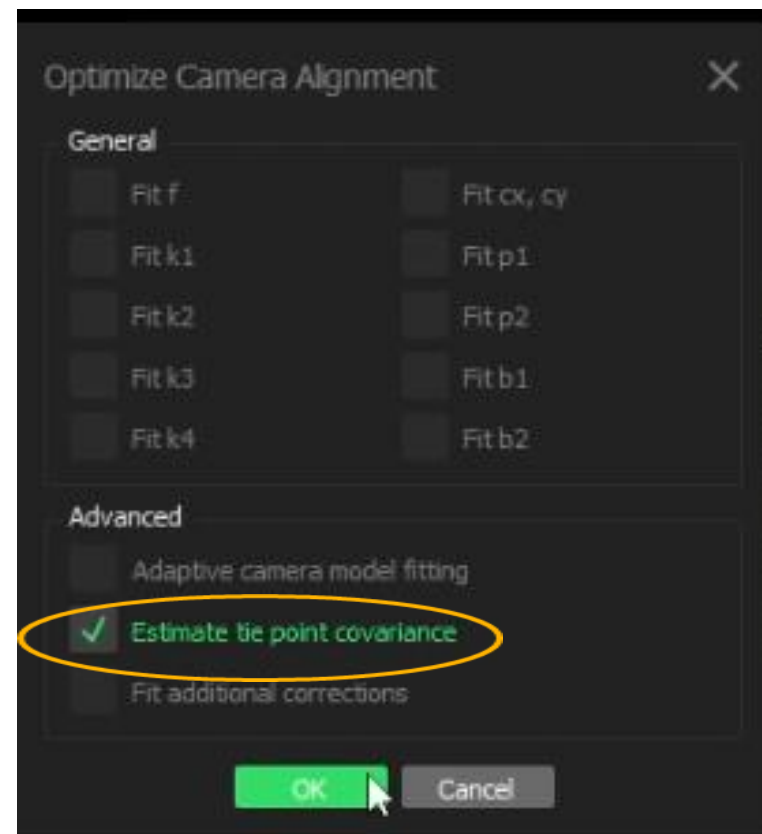
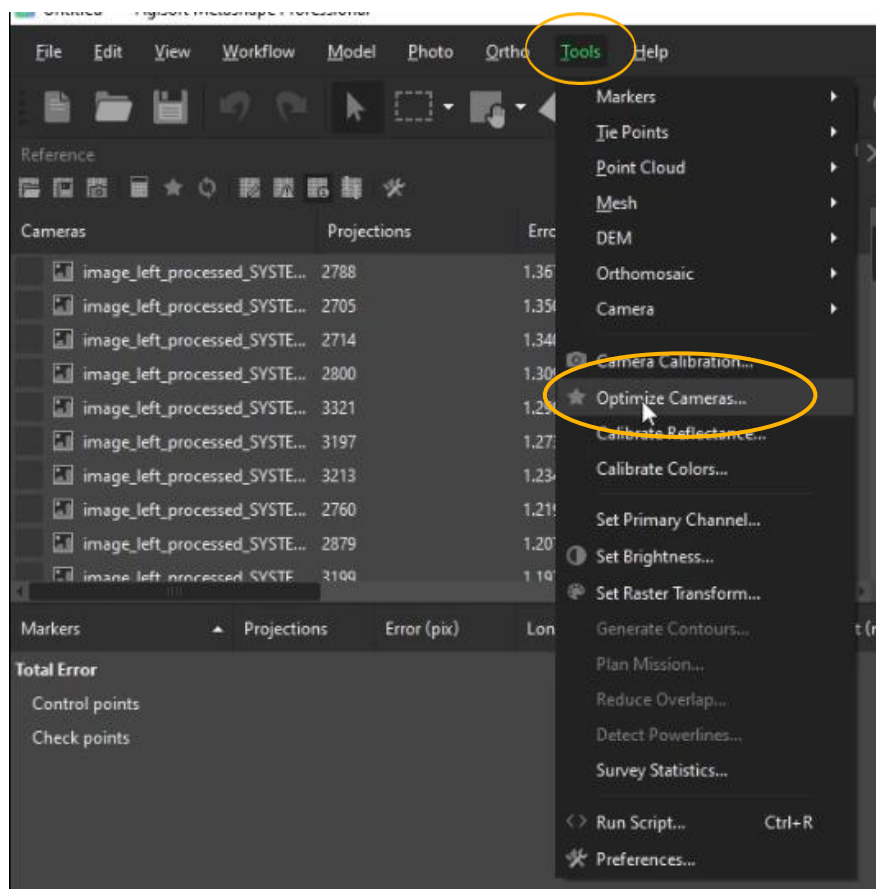
Once your stills have been aligned. You may need to click Reset the View in order to find your dataset.





# Optimize Camera Alignment

Tools → Optimize Cameras → Select “Estimate tie point covariance”



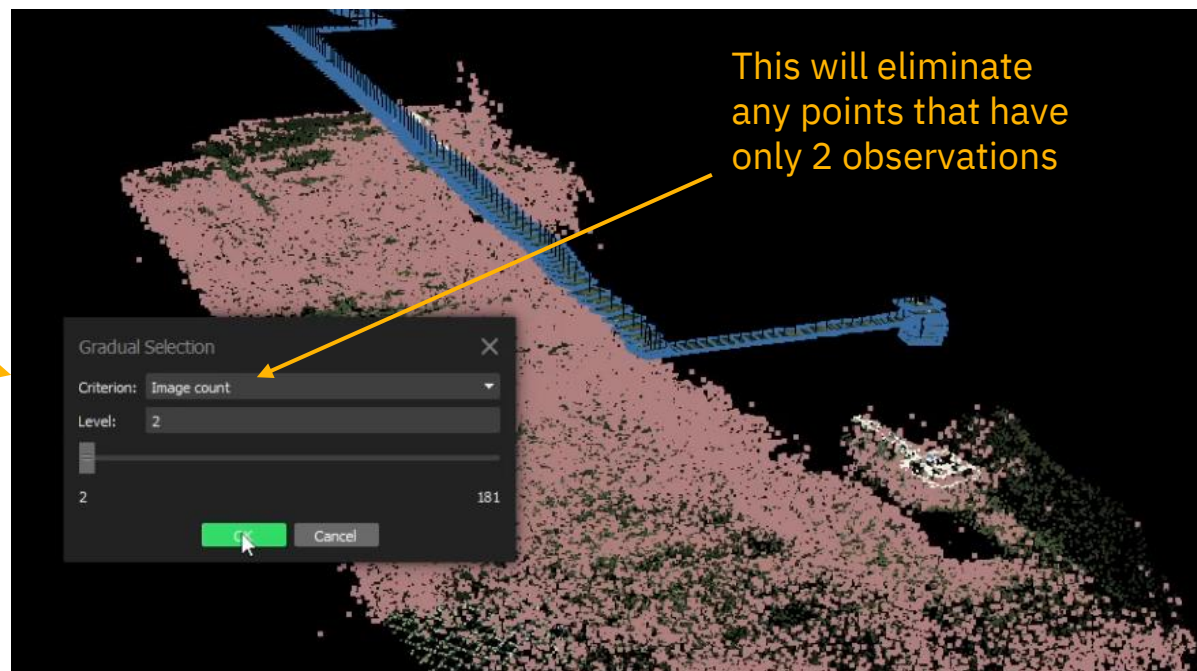
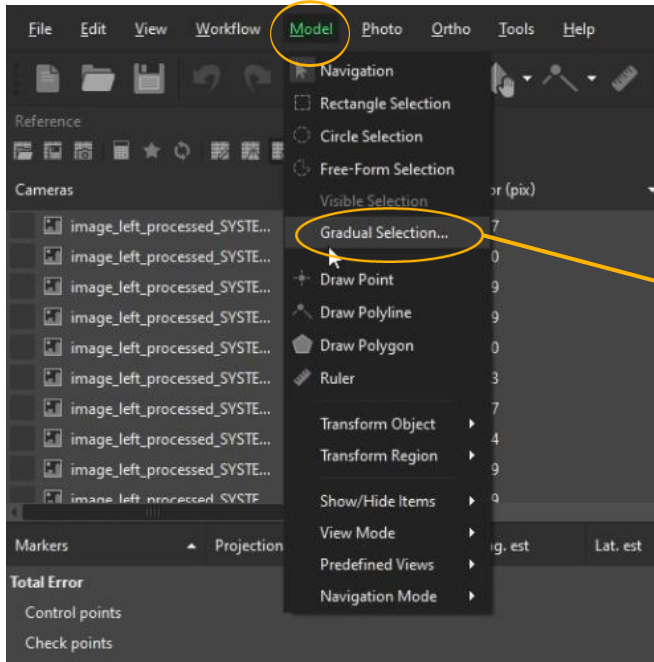


# Filter: Key Point Features

This step allows the user to remove low-quality **key point features** from your dataset (removing key points)

- These need to be eliminated to have a successful mesh model
- If the “Feature” is only seen from 2 perspectives, then it should be deleted
  - These features typically have a high uncertainty associated with them

Remove these points: **Model → Gradual Selection → Image Count, Level = 2 → OK → Delete**



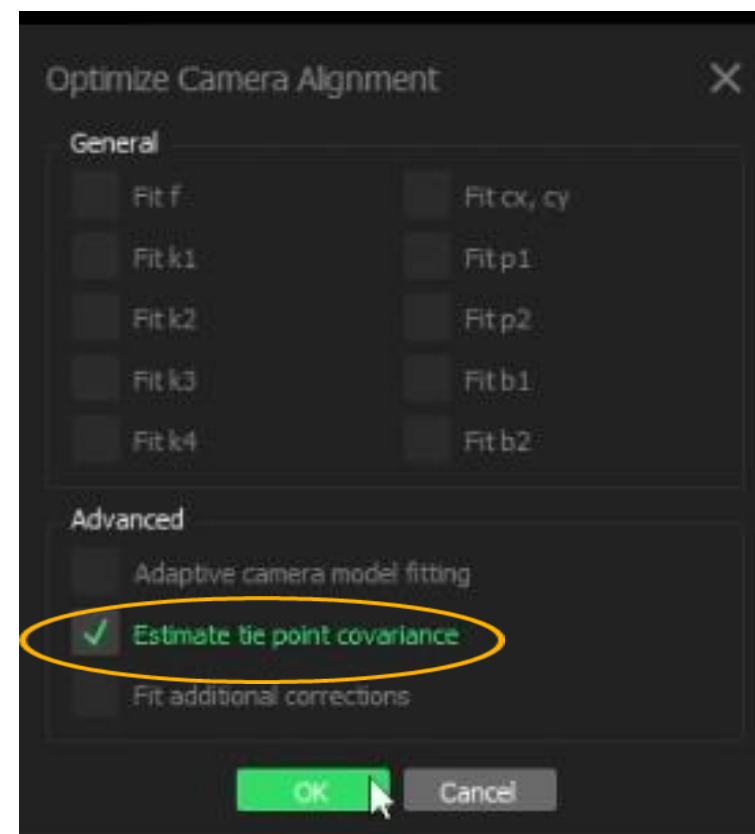
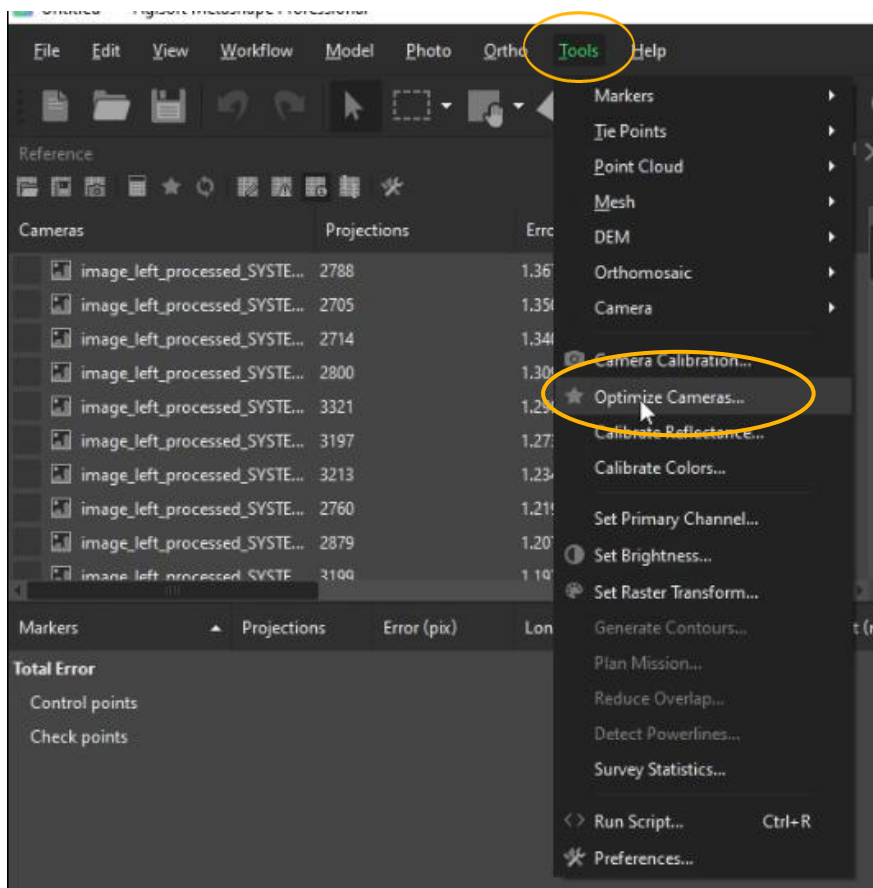




# Optimize Camera Alignment

Every time you filter your dataset (Ex. Step 8) you need to **Optimize Cameras Again**

Tools → Optimize Cameras → Select Estimate tie point covariance → OK





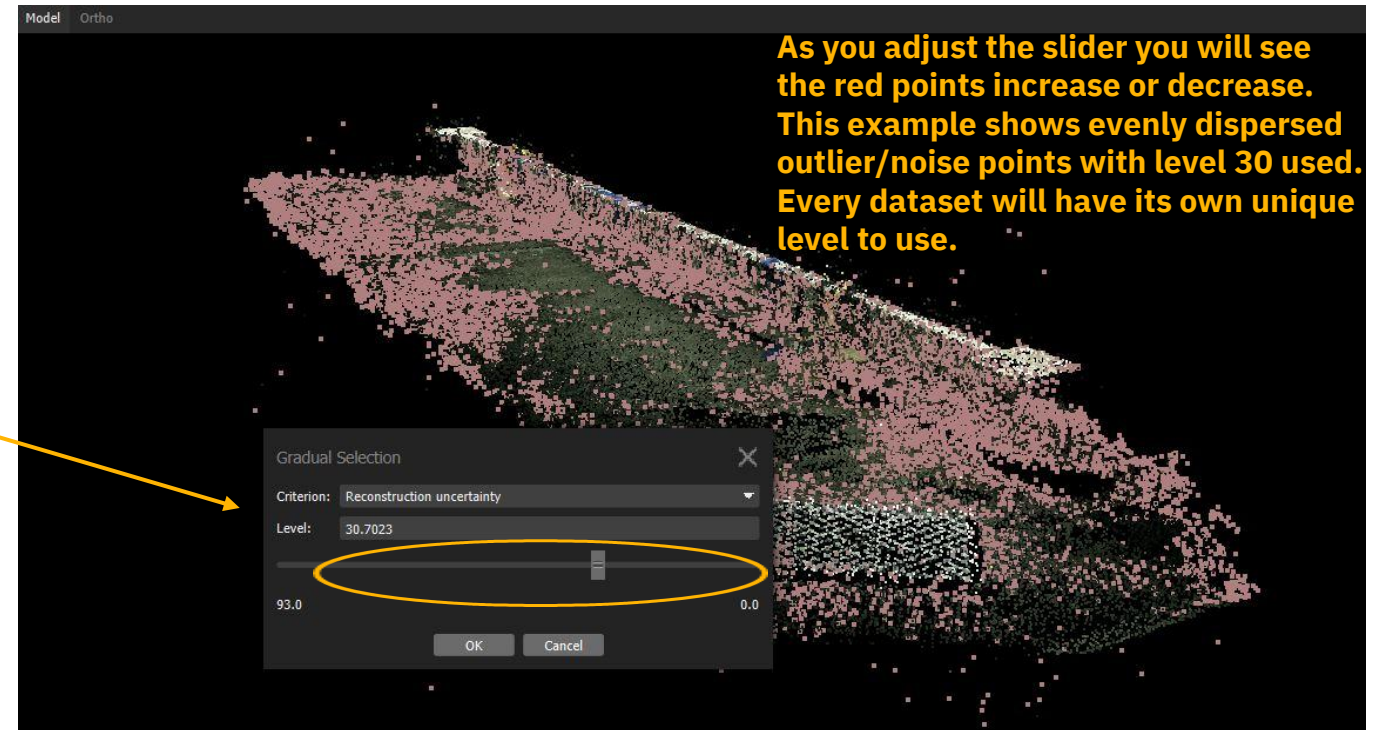
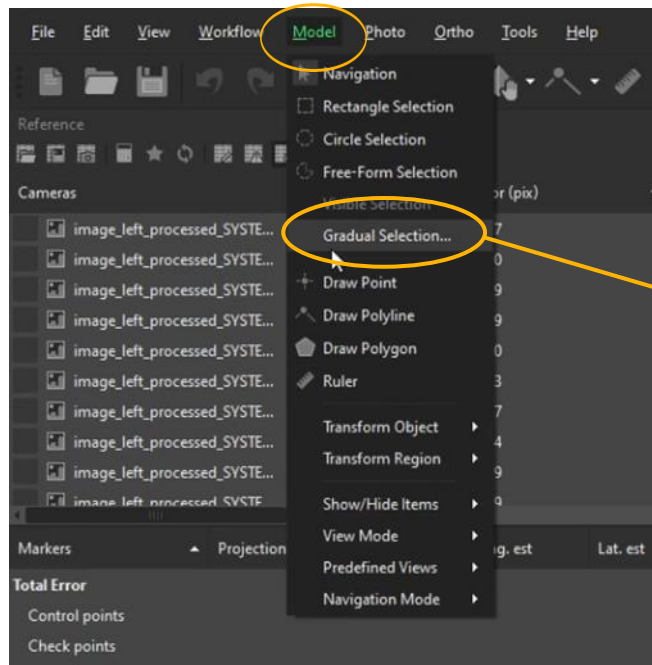
# Filter: Reconstruction Uncertainty

An additional filtering step. Use the Slider to adjust the **Reconstruction Uncertainty Level** until you get an even level of outlier tie points (noisy points likely not associated with solid objects).

- Play with the slider until the Pink Dots are evenly distributed across the model's noise

**Model → Gradual Selection → Reconstruction Uncertainty → Level TBD → OK → Delete**

- \*\*\* You must click “Delete” on your keyboard to delete the selected points



**Optimize Cameras Again:** Tools → Optimize Cameras → Estimate Tie point covariance → OK

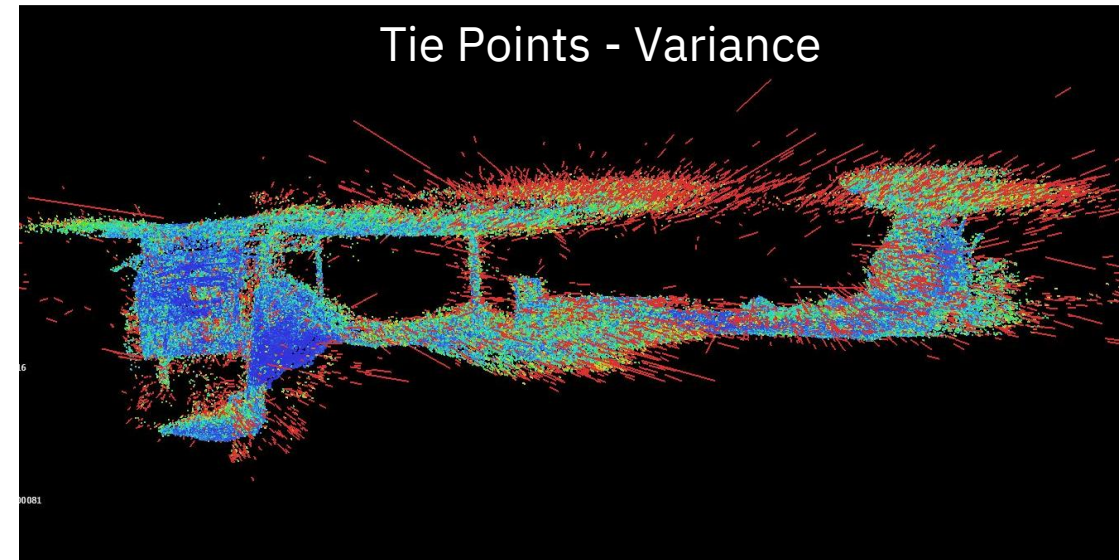
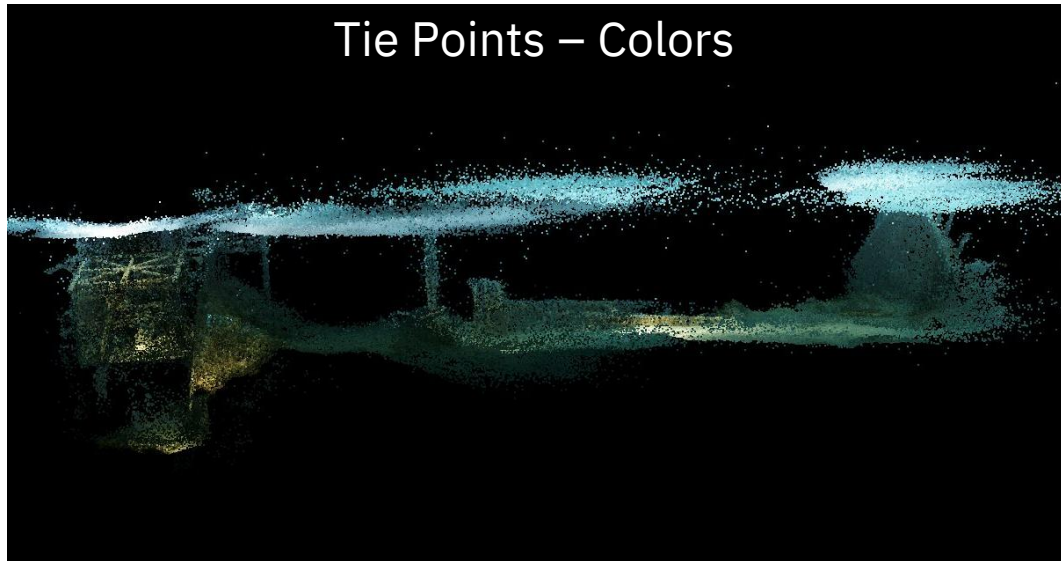
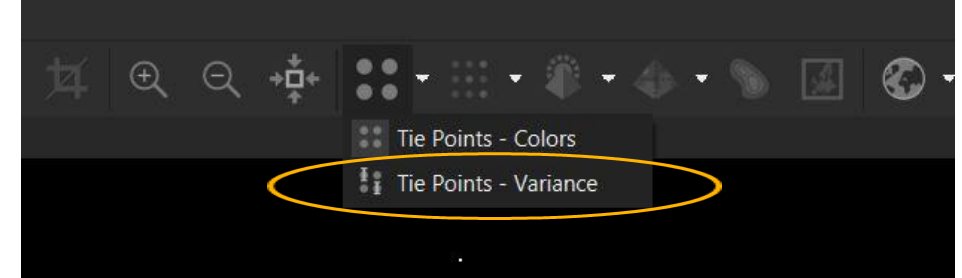
# Visualizing Tie Points Variance

Select the menu icon shown to view “Tie Points – Variance”.

**Tie Points** are the “connections” between sequential images in the survey and connect the model together.

**This visualizes the uncertainty associated with each data point. Long red lines are highly uncertain and often outliers.**

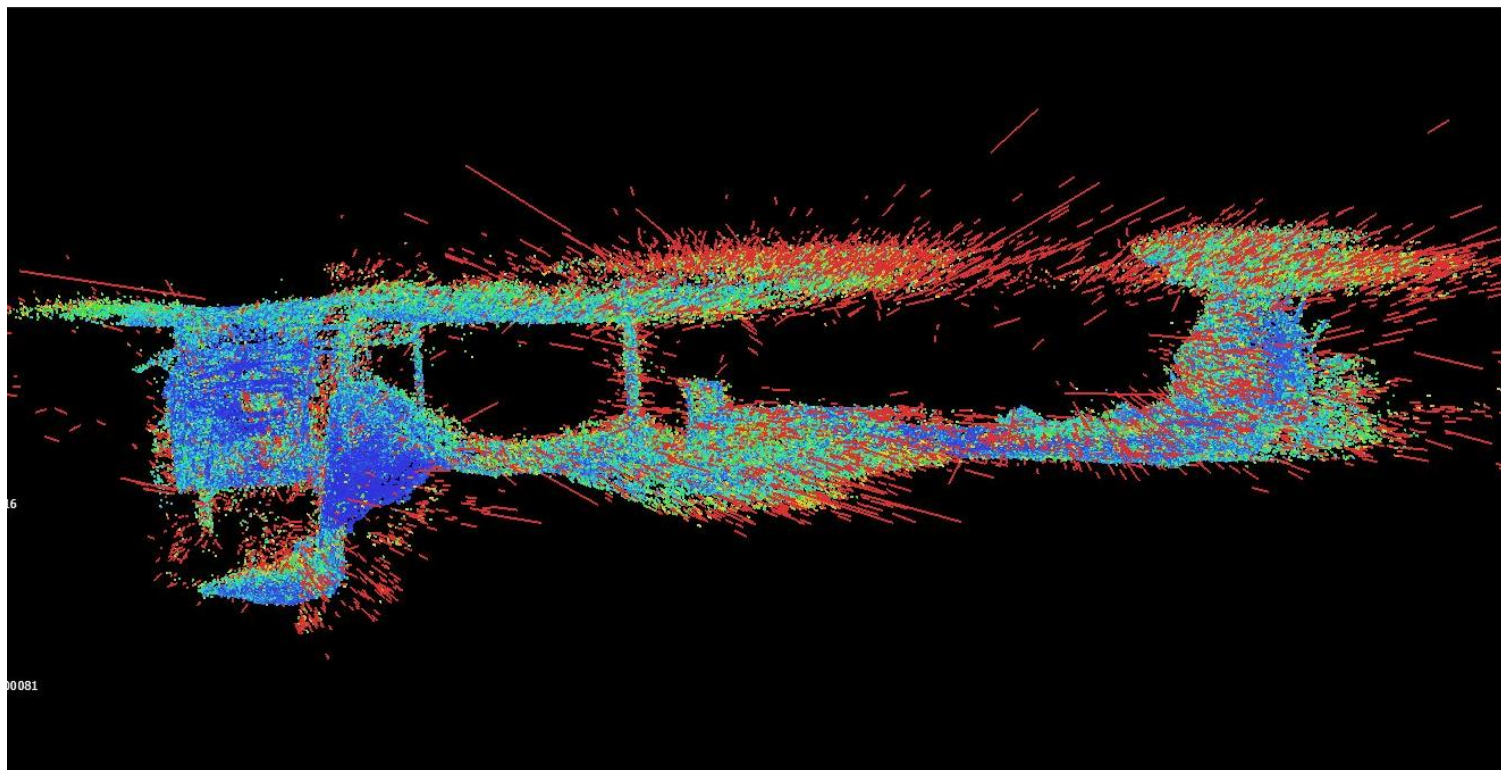
When turned on, you’ll be looking for any red spikes in your dataset. The next step will help remove them.





# Visualizing Tie Points - Variance

Tie Points - Variance



The **blue highlights** are where the higher quality image alignment is.

The **red lines** show where the higher uncertainty is.

This will illustrate where confident tie points exist and where uncertainty is present.



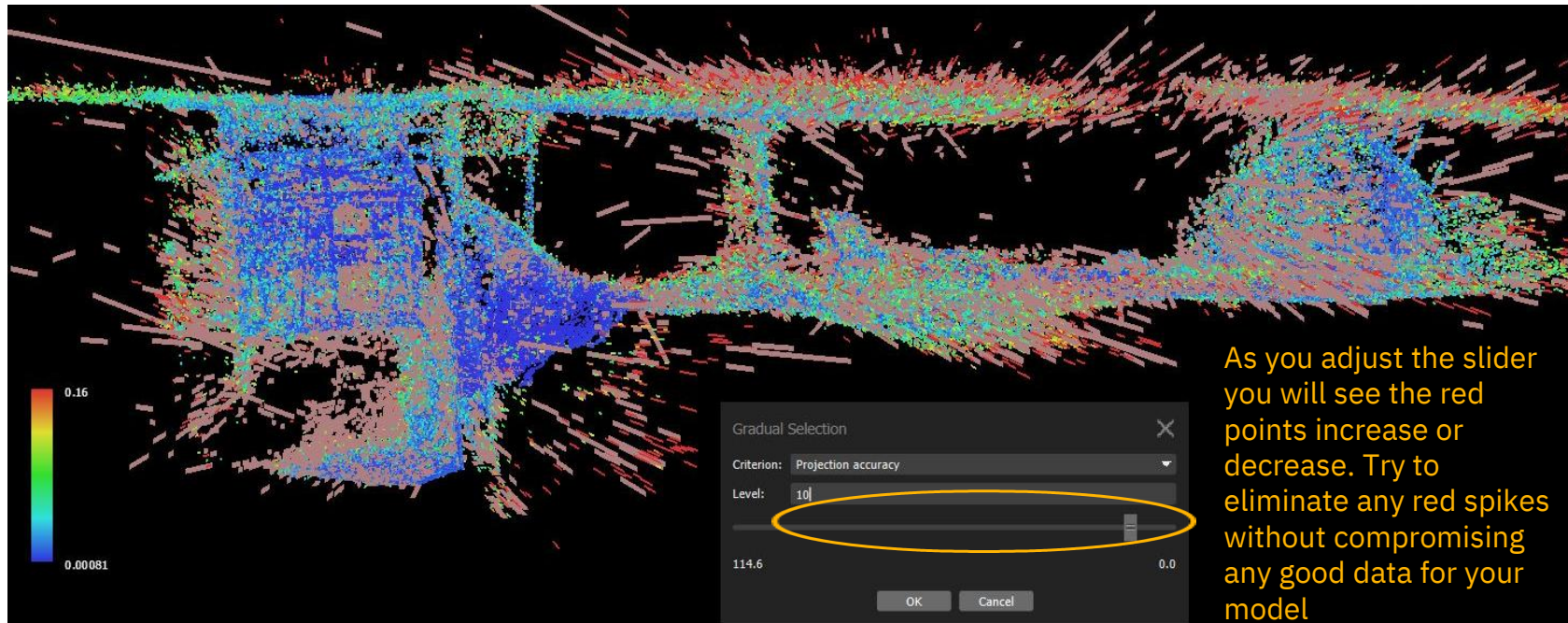


# Filter: Projection Accuracy

Gradual Selection Filter for **Position Accuracy** will remove low quality tie points in your dataset (red lines)

- Use slider optimize removal – Do this iteratively, starting small so you don't remove too many points
- 10 often works well, but tune to your project for best results

**Model → Gradual Selection → Projection Accuracy → Select Level (Ex. 10) → OK → Delete on Keyboard**



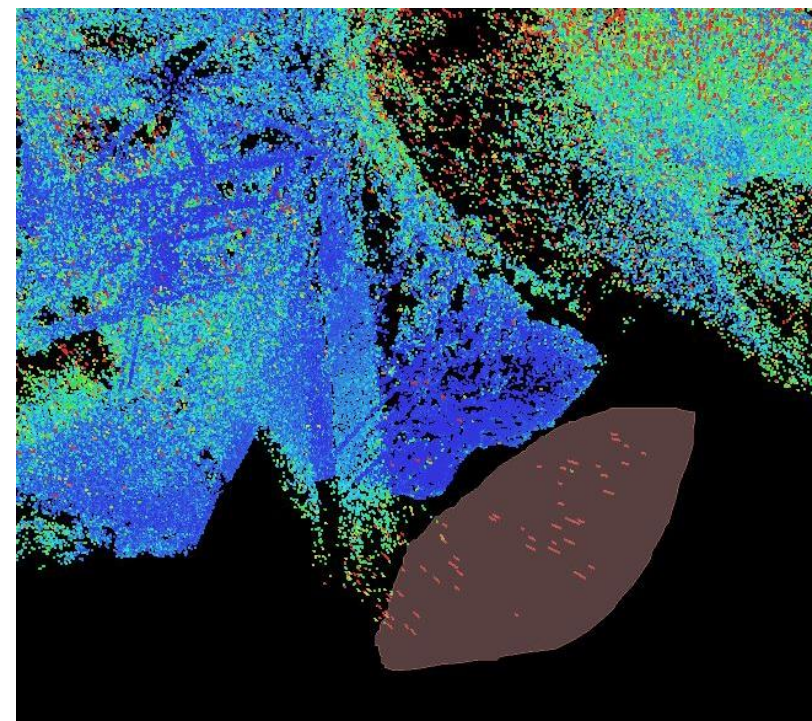
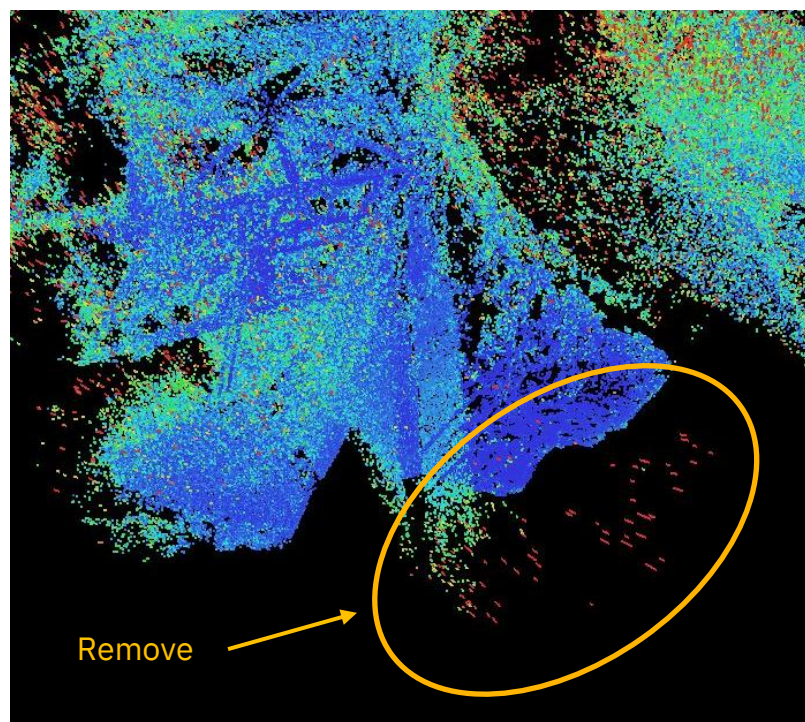
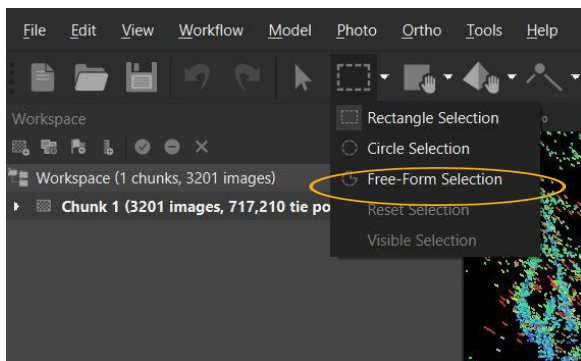
**Optimize Cameras Again:** Tools → Optimize Cameras → Estimate Tie point covariance → OK



# Free Form Select

Free-Form Selection Tool allows the user to manually remove any outliers that you find “sitting in space” which could not be removed from the Gradual Selection Filter process.

Simple manual tool where you “Lasso” around the outlier points to **Select** → DEL on keyboard to **Remove**



**Optimize Cameras Again:** Tools → Optimize Cameras → Estimate Tie point covariance → OK



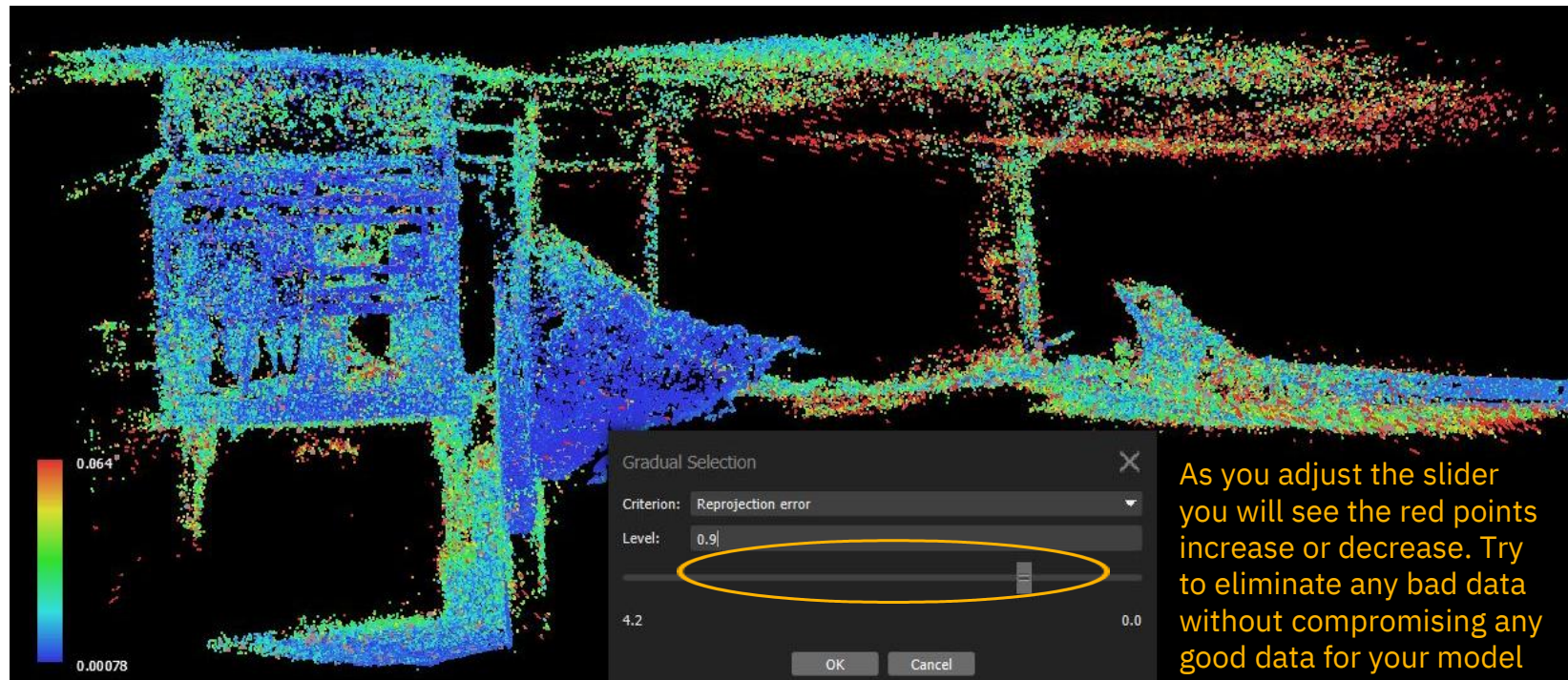


# Filter: Reprojection Error

This Gradual Selection for Reprojection Error will remove data points with high reprojection error

- Very dependent on the dataset – Iterative and observe what it removes in the model
- Generally “0.9” or lower seems to work well

**Model → Gradual Selection → Reprojection Error → 0.9 → OK → DEL**



**Optimize Cameras Again:** Tools → Optimize Cameras → Estimate Tie point covariance → OK



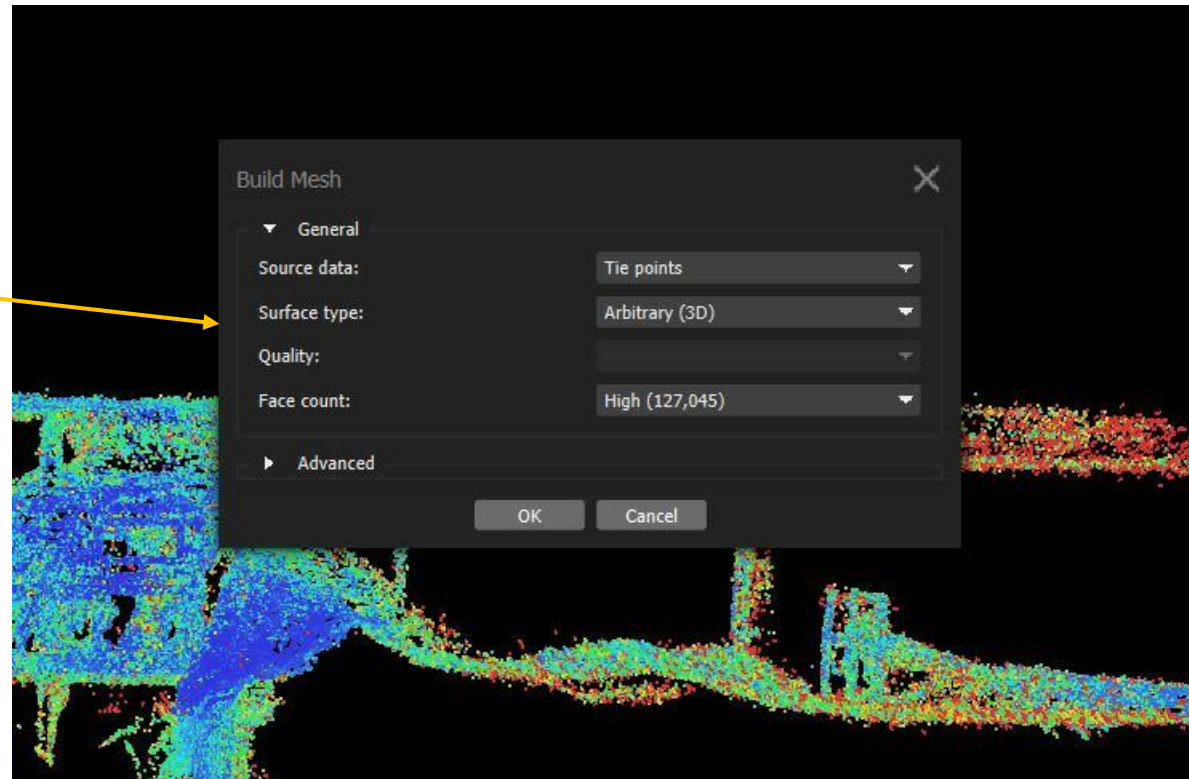
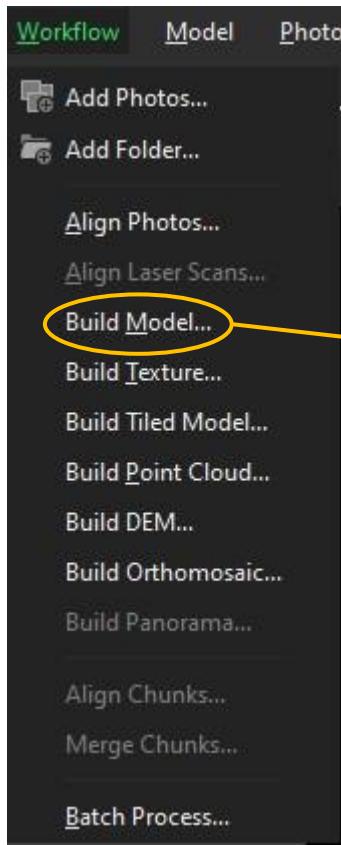
# Build Model



# Build Model Settings

Once you have aligned and optimized your model, now generate an initial 3D Mesh  
**Workflow → Build Model**

Source Data = Tie Points  
Surface Type = Arbitrary  
Face Count = High



# Build Model Settings

Once your mesh has been created, you will need to find the model in the tree.

First, click **Workspace**, then double-click on **3D Model**



\*Note, the output will not look ideal, this is just the beginning of the mesh building process.\*



# Build Model

Once you have completed “Reduced Overlap”, you are now ready to build your final 3D Mesh Model.

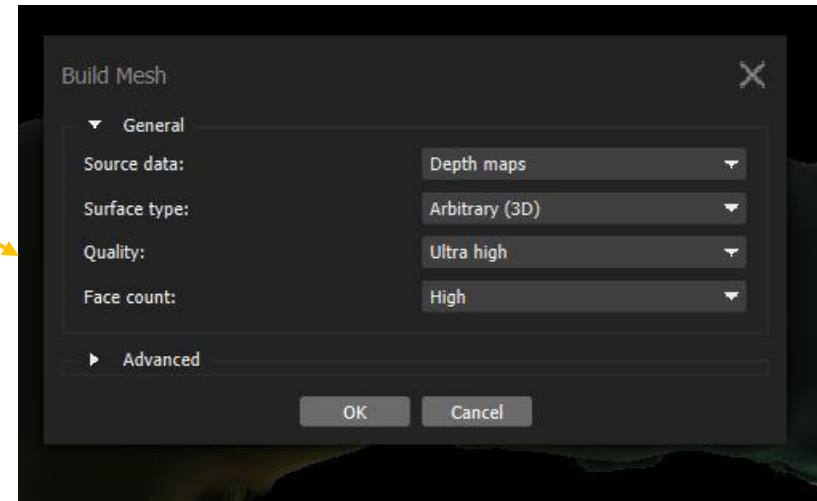
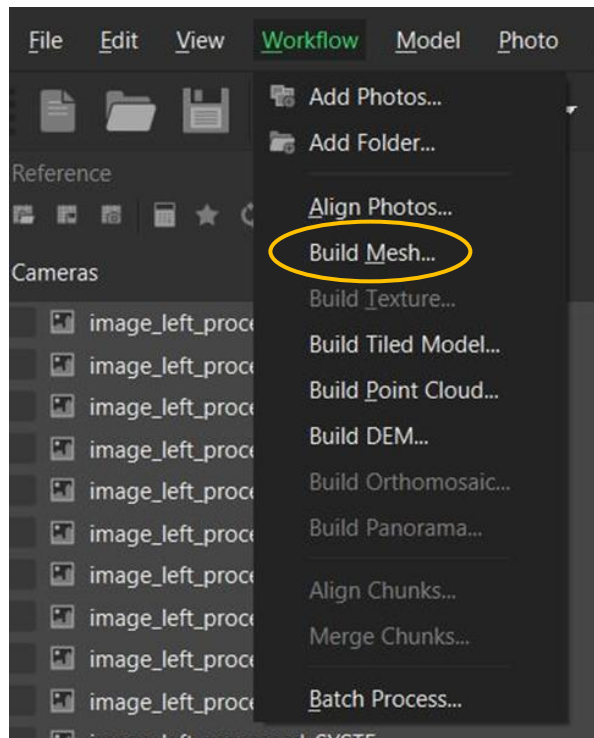
**Workflow → Build Model**

**Source Data** = Depth Maps (was tie points for initial mesh)

**Surface Type** = Arbitrary

**Quality** = Ultra High

**Face Count** = High





BUILD MODEL



# Model Result





# Filter: Components

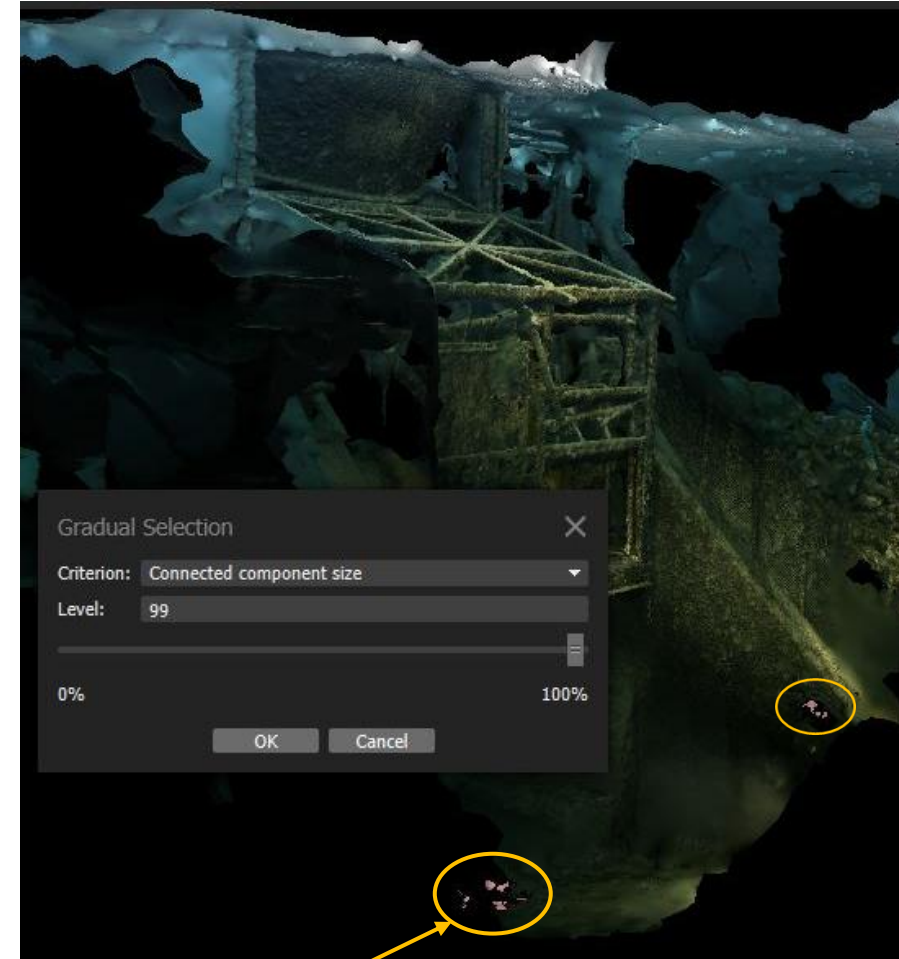
If there are pieces of mesh that exist outside of your main object, you can quickly clean it

**1. Model → Gradual Selection → Connected Component Size → “N” → OK**

- Setting the Level to “99” will result in a single component
- Adjust to remove obvious outlier mesh components. You will want to keep multiple components if your model is not fully connected.

**2. If the floating unimportant piece(s) are highlighted (they will turn pink), press the delete key on the keyboard to remove them.**

**Note:** You can also use the Selection Tools to manually select and delete unimportant areas.



*Highlighted* areas that will be deleted

# Texture



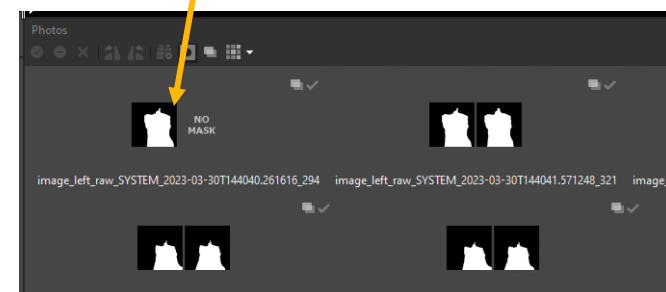
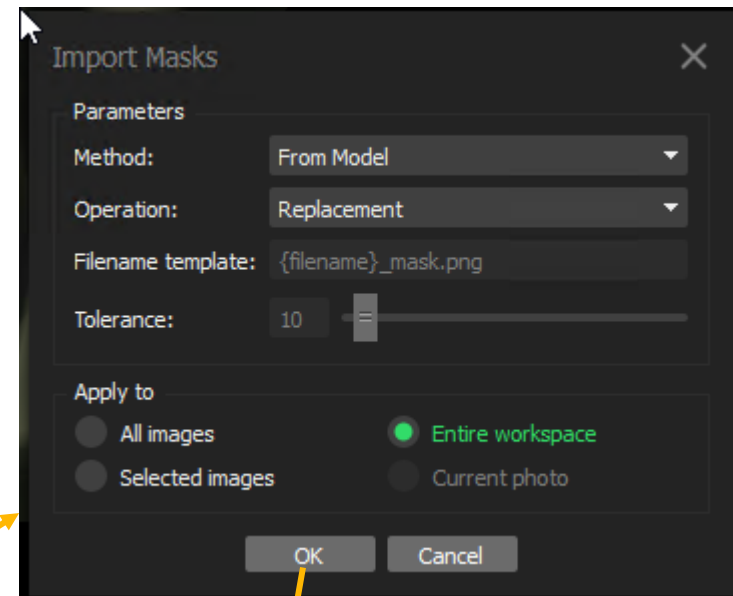
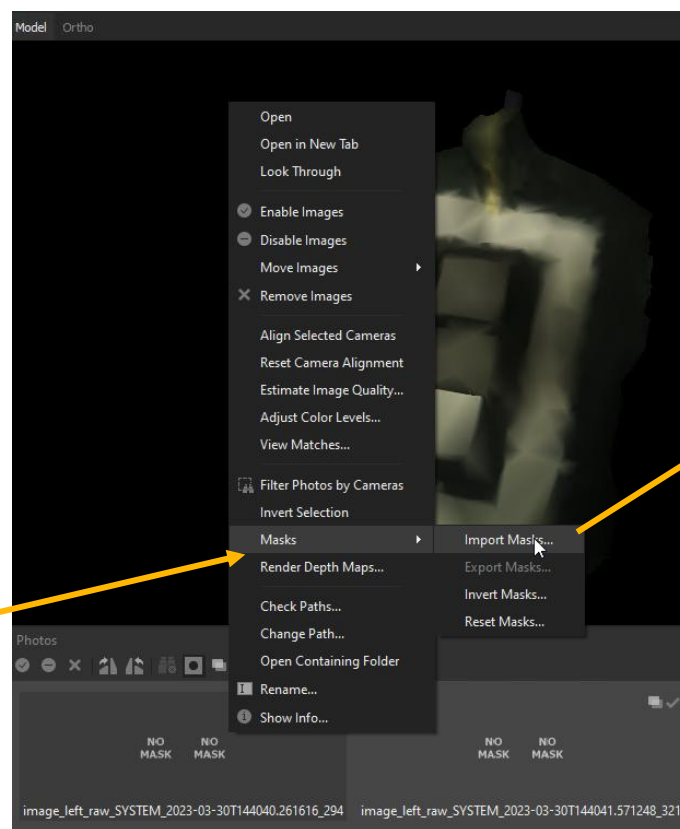
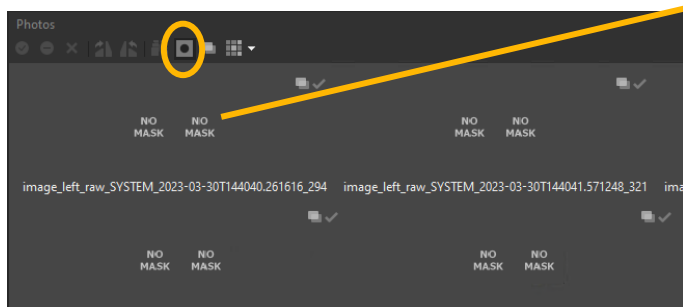


## BUILD MODEL

# Masking

Masking determines which images to use for the texturization process

1. Go to **Photos** pane
2. Click on **“Show Masks”**
3. You should see NO MASKS to start
4. Right Click any “NO MASK” image:
  - i. Go to **Masks** → **Import Masks**
    - i. Method: **From Model**
    - ii. Operation: **Replacement**
    - iii. Apply to: **Entire Workspace**
    - iv. Click **“OK”**
5. Some images should now be used as masks



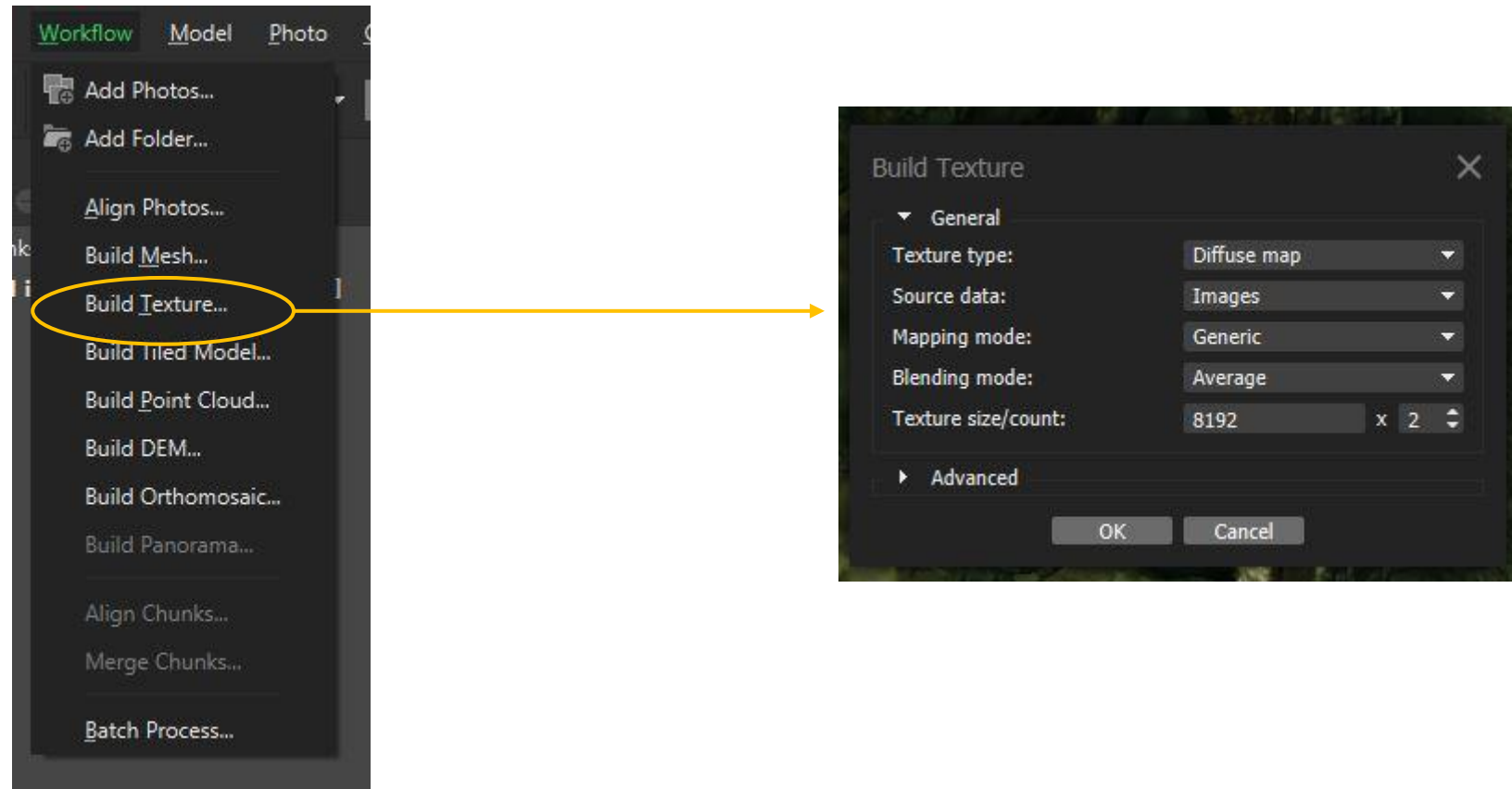


# Build Texture

Once your Mesh model has been created you can add texture to it.

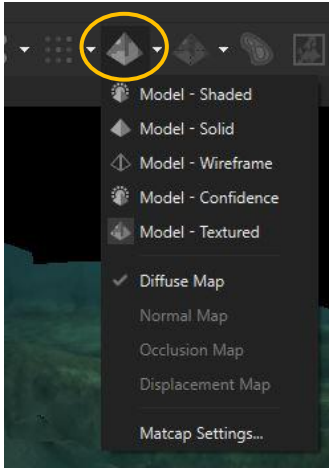
**Workflow → Build Texture**

Use Default settings and Click “**OK**”



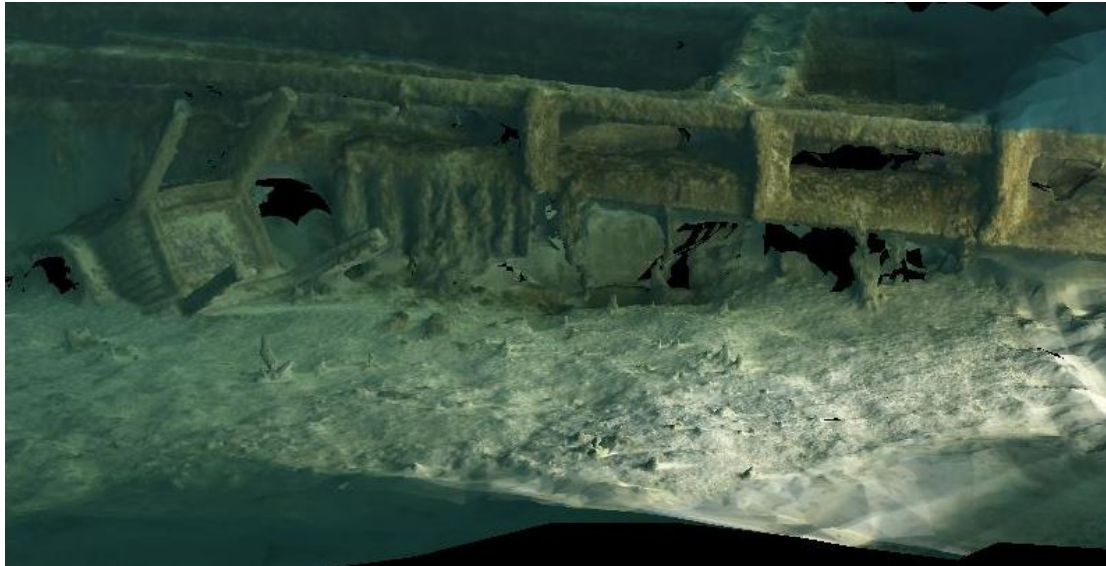


# Model – Textured or Shaded



Once a textured model is created, you will have the option to view different models. Click the **Prism** icon and select which model you want to view.

**Note: “Model – Shaded” can look better than “Model – Textured”**



Model - Textured



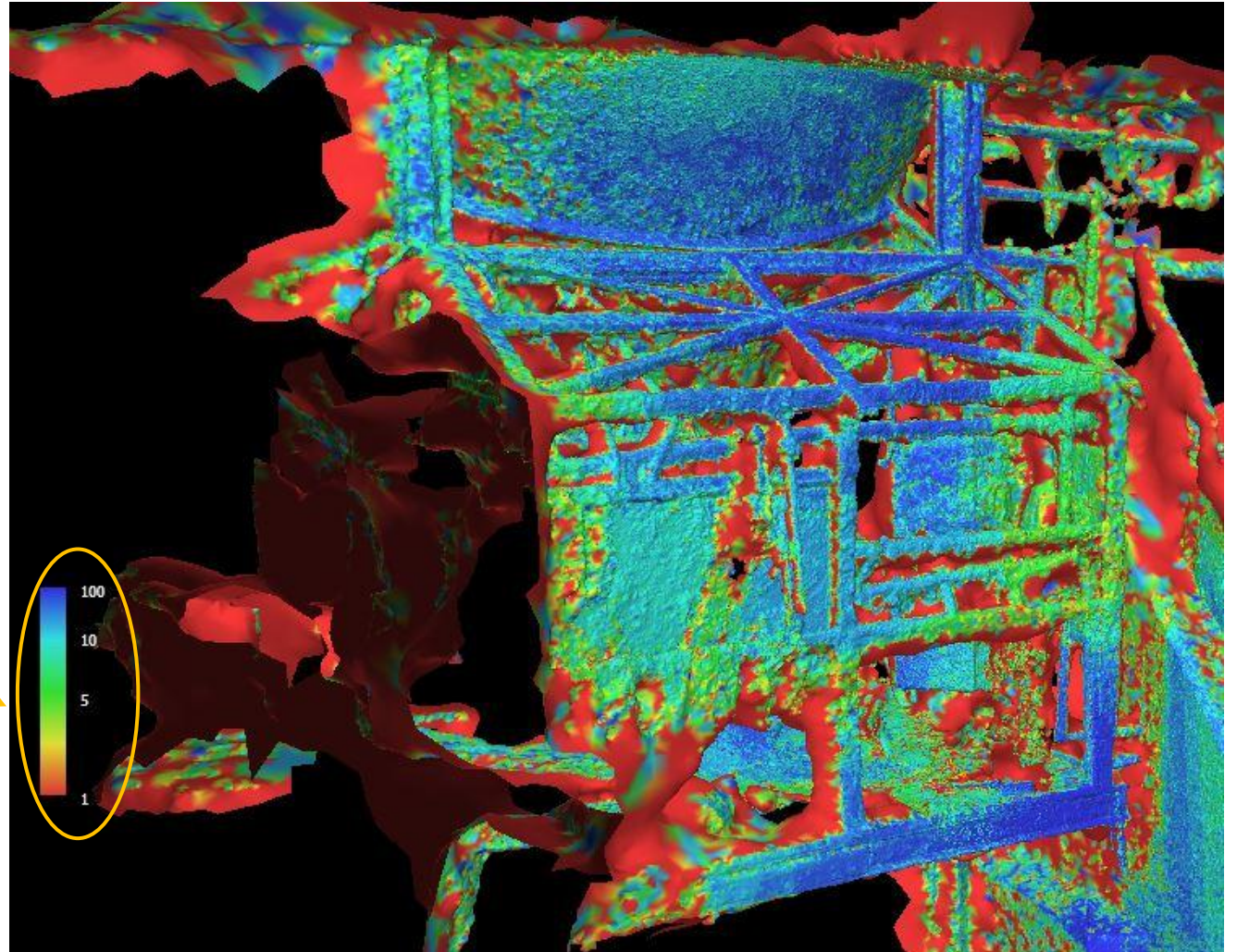
Model - Shadowed



# Model – Confidence

“**Model – Confidence**” will allow the user to see which areas of their model have the most “certainty”

Blue = High Confidence  
Red = Low Confidence



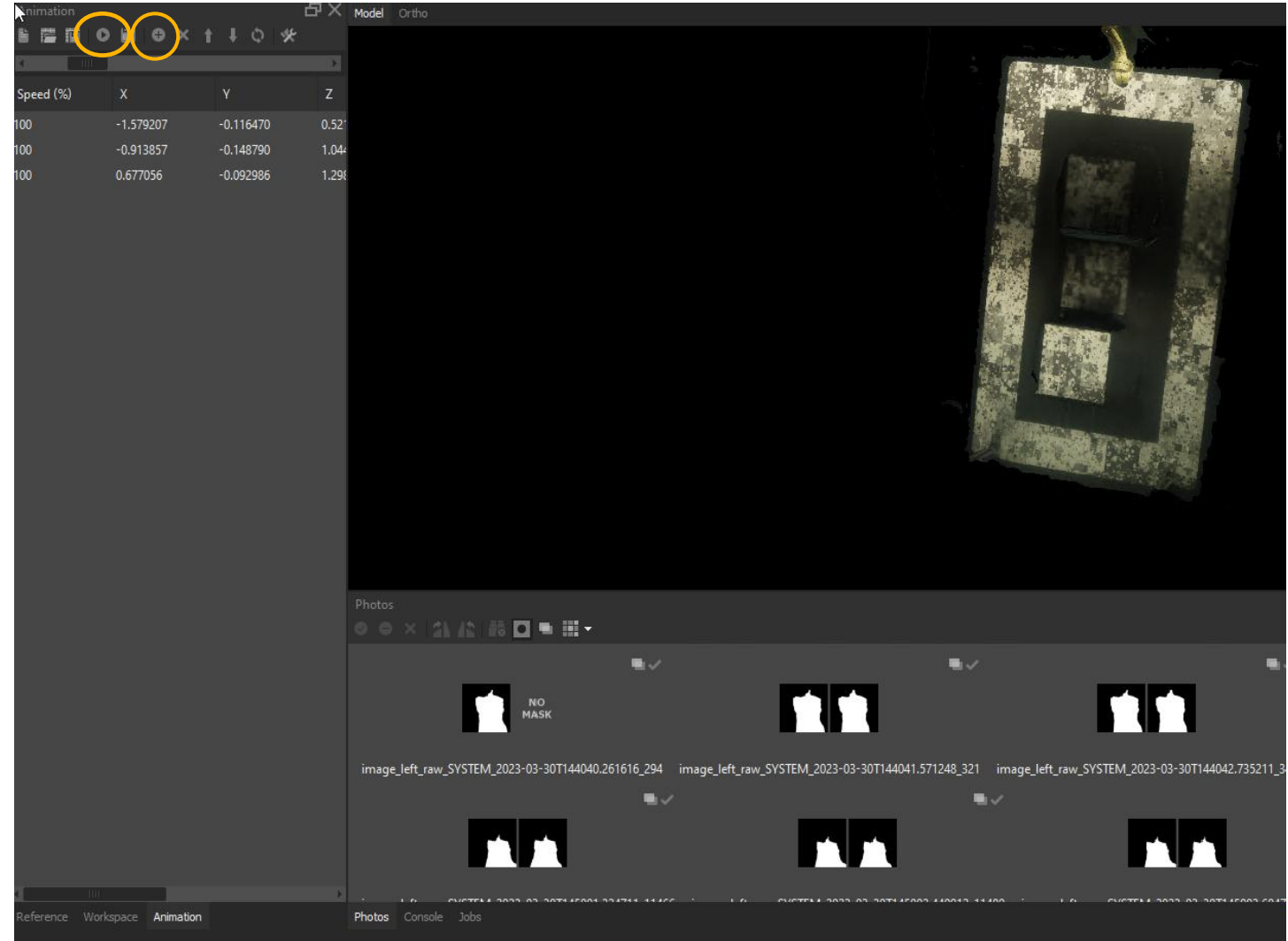
# Animate





# Animate a Video

- Go to **View** → **Animation**
  - Rotate and move your model to create Viewpoints. The “camera” will fly through these perspectives to create the video.
  - Click the “**Append**” button to make a new viewpoint, **Plus** icon. 
  - Add as many viewpoints as you want in your video.
  - Preview the video by clicking the **Play** icon and change playback settings with the “**Settings**”, tool icon
  - Once ready, click “**Capture**”, use the floppy disk icon to save the video.



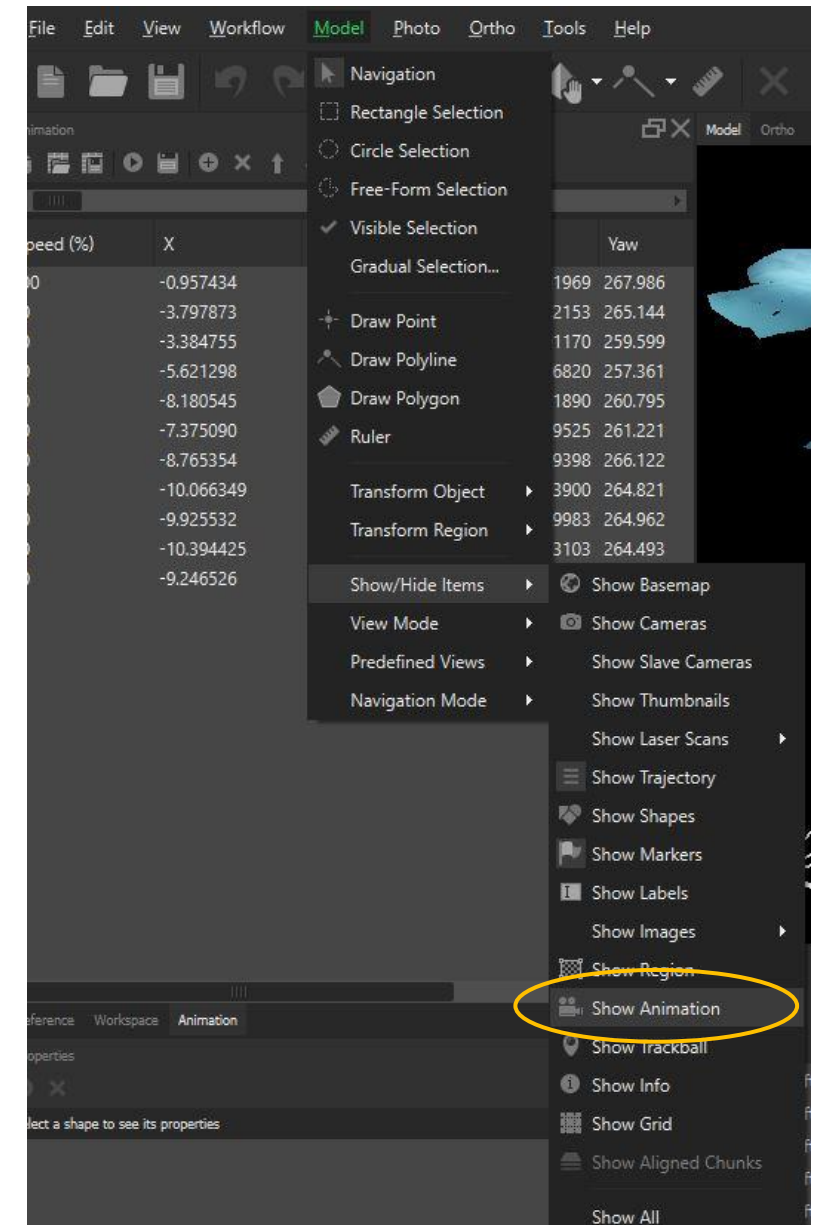
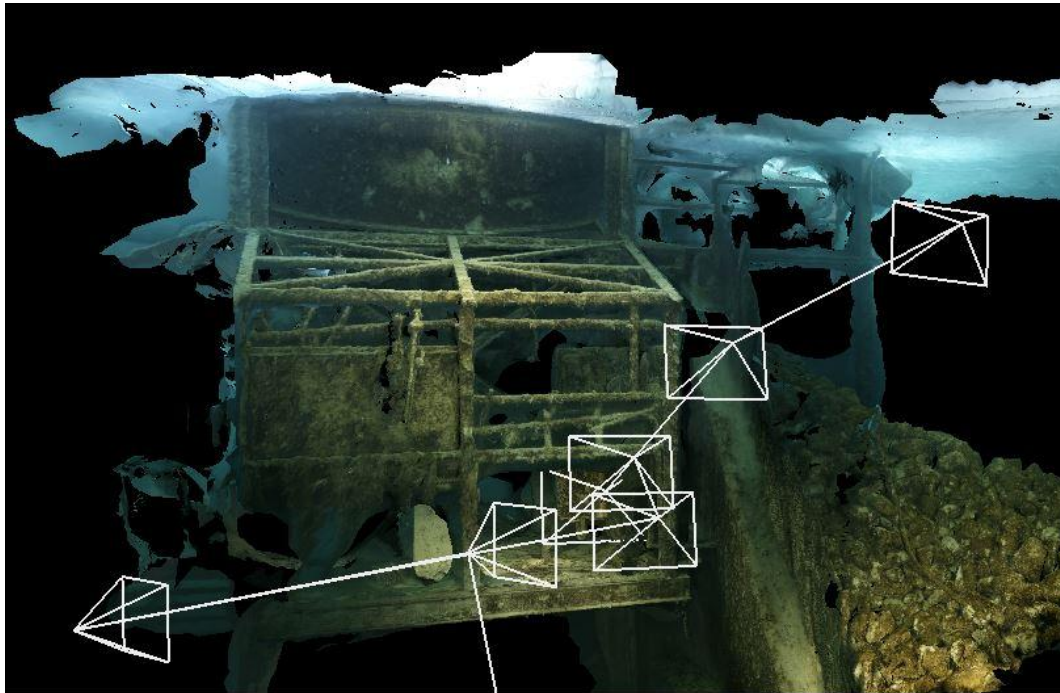


# Animation

Once you have created an animation file, you will see white camera animation symbols in your video.

These can be turned off in the settings

Model → Show/Hide Items → Show Animation



# Exports







# Exporting Options

- To export your Model, simply go to **File → Export**.
- You'll be presented with a variety of formats to choose from.
- In order to export a point cloud, it is necessary to create one beforehand:
  - **Workflow → Build Point Cloud**
  - Once built you will be able to export in the export settings

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